Supplementary Materials
HypEReg-TransMorph: A Hyperelastic-Regularized Transformer Framework for Folding-Suppressed Deformable Brain MRI Registration

This document accompanies the manuscript *HypEReg-TransMorph: A Hyperelastic-Regularized Transformer Framework for Folding-Suppressed Deformable Brain MRI Registration.* Extended quantitative results, inferential tables (S1–S2), grouped-VOI mapping (S3), Figure S1, and implementation/profiling details are provided here as separate Supplementary Materials (they are not duplicated in the main journal PDF).

1 Supplementary Experimental Details

1.1 Model and Environment Configuration

This supplementary document summarizes implementation details required for reproducibility:

- Framework and package versions: Python 3.12.10; PyTorch 2.12.0.dev20260408+cu128; antspyx 0.6.3; SimpleITK 2.5.3.
- Hardware configuration: NVIDIA RTX PRO 6000 Blackwell Workstation Edition (97,887 MiB VRAM), driver 591.86; peak-memory audits from per-case evaluation logs indicate a single-GPU run profile for deep models.
- Checkpoint management follows a unified manifest-and-version protocol aligned with the benchmark settings.
- Inference is conducted under a standardized evaluation configuration with harmonized metric definitions across models.
- Isolated forward profiling is executed in a clean no-gradient setting without label-warp or metric kernels. Measured pure-forward runtime/peak-memory for HypEReg-TransMorph and TransMorph are 0.4316 s / 5.422 GB and 0.4306 s / 5.421 GB, respectively.

2 Figure S1: Descriptive IXI Overview

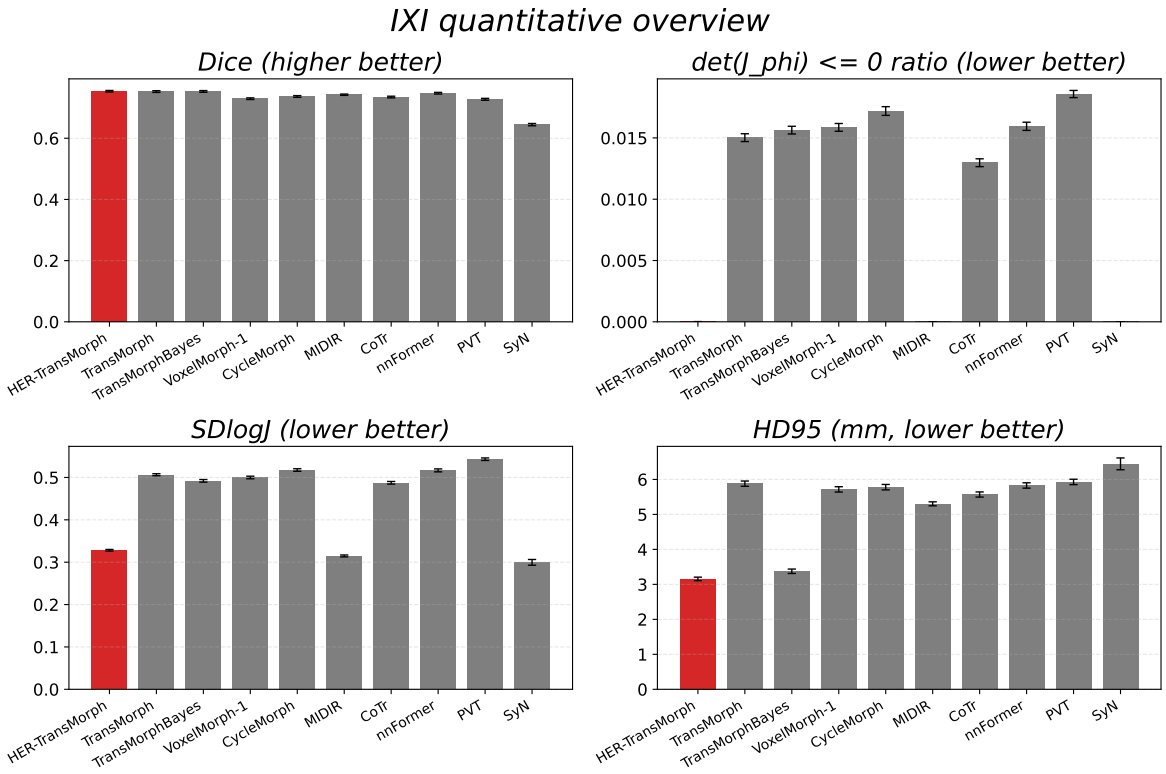

Figure S1: Descriptive IXI quantitative overview (Dice, non-positive Jacobian determinant ratio, SDlogJ, and HD95). Bars include standard-error error bars for visual uncertainty context.

3 Inferential Statistics (HypEReg-TransMorph vs. Baselines)

Table S1: Inferential summary for TransMorphBayes and MIDIR (mean±std, aligned median difference, Wilcoxon p, Benjamini–Hochberg q, and matched-pairs rank-biserial effect). HypEReg-TransMorph denotes the proposed model.

Metric	**Baseline**	**HypEReg mean±std**	**Baseline mean±std**	**Median (aligned)**	**diff**	p/q**/effect**
dice_mean	TransMorphBayes	0.5776±0.0303	0.5743±0.0315	0.00249		6.57×10^{-8} / 7.58×10^{-8} / 0.6522
non_jec	TransMorphBayes	$1.39 \times 10^{-5} \pm 7.55 \times 10^{-6}$	0.01484±0.00322	0.01449		1.19×10^{-16} / 1.99×10^{-16} / 1.0000
SDlogJ	TransMorphBayes	0.3268±0.0216	0.4908±0.0313	0.16264		1.19×10^{-16} / 1.99×10^{-16} / 1.0000
HD95_mean	TransMorphBayes	5.3032±0.7228	5.6881±0.7697	0.33877		1.34×10^{-14} / 1.88×10^{-14} / 0.9298
ASSD_mean	TransMorphBayes	1.3559±0.1809	1.4110±0.1839	0.05799		4.12×10^{-8} / 4.88×10^{-8} / 0.6622
dice_mean	MIDIR	0.5765±0.0301	0.5643±0.0244	0.01166		3.79×10^{-18} / 7.76×10^{-18} / 0.9331
non_jec	MIDIR	$1.46 \times 10^{-5} \pm 7.37 \times 10^{-6}$	0.0000±0.0000	-1.31×10^{-5}		1.31×10^{-20} / 3.69×10^{-20} / -1.0000
SDlogJ	MIDIR	0.3280±0.0221	0.3148±0.0242	-0.01365		4.84×10^{-20} / 1.28×10^{-19} / -0.9850
HD95_mean	MIDIR	5.3234±0.6936	5.3028±0.6139	-0.00855		0.5767 / 0.5898 / -0.0600
ASSD_mean	MIDIR	1.3570±0.1770	1.4100±0.1585	0.06514		3.58×10^{-9} / 4.36×10^{-9} / 0.6342

Aligned median difference is oriented so positive values indicate HypEReg-TransMorph improvement for the given metric direction. The effect is matched-pairs rank-biserial correlation. Pair counts differ by baseline (TransMorphBayes: $n = 91$; MIDIR: $n = 115$).

Table S2: Selected paired comparisons (Wilcoxon signed-rank with Benjamini–Hochberg FDR). Positive signed effect indicates HypEReg-TransMorph improvement under metric-aligned directionality; signed effect is matched-pairs rank-biserial correlation.

Metric	**Baseline**	**Median paired diff (aligned)**	**p-value**	**q-value**	**Signed effect**
non_jec	TransMorph	0.014747	1.31×10^{-20}	3.69×10^{-20}	1.0000
SDlogJ	TransMorph	0.170655	1.31×10^{-20}	3.69×10^{-20}	1.0000
HD95_mean	TransMorph	0.319583	3.77×10^{-17}	7.08×10^{-17}	0.9046
ASSD_mean	TransMorph	0.051747	6.09×10^{-11}	7.83×10^{-11}	0.7028
non_jec	MIDIR	-0.000013	1.31×10^{-20}	3.69×10^{-20}	-1.0000
SDlogJ	MIDIR	-0.013652	4.84×10^{-20}	1.28×10^{-19}	-0.9850
HD95_mean	MIDIR	-0.008554	0.5767	0.5898	-0.0600
ASSD_mean	MIDIR	0.065138	3.58×10^{-9}	4.36×10^{-9}	0.6342
non_jec	TransMorph (uploaded ckpt)	0.014747	1.31×10^{-20}	2.63×10^{-20}	1.0000
dice_mean	TransMorph (uploaded ckpt)	0.002326	6.88×10^{-7}	6.88×10^{-7}	0.5334
non_jec	TransMorphBayes (uploaded ckpt)	0.014714	1.31×10^{-20}	2.63×10^{-20}	1.0000
dice_mean	TransMorphBayes (uploaded ckpt)	0.002706	1.74×10^{-10}	2.32×10^{-10}	0.6858
non_jec	SyN	-0.000012	8.00×10^{-16}	1.24×10^{-15}	-0.8654
SDlogJ	SyN	-0.045599	3.09×10^{-5}	3.31×10^{-5}	-0.4477
HD95_mean	SyN	0.705730	1.31×10^{-7}	1.47×10^{-7}	0.5670
ASSD_mean	SyN	-0.111801	0.6572	0.6572	0.0477

Full inferential outputs across all exported metric-baseline pairs are provided in the supplementary release package. Additional uploaded-checkpoint paired results for TransMorph and TransMorphBayes (Dice and non-positive Jacobian ratio) are included with corresponding subject-level inputs. Repeated values such as $p = 1.31 \times 10^{-20}$ occur when the two-sided Wilcoxon test reaches the practical floating-point floor under near-complete sign consistency for $n = 115$, and should be interpreted as extremely small p-values.

4 Table S3: Grouped-VOI Mapping

Table S3: Grouped-VOI mapping used for grouped Dice reporting.

Grouped VOI	**FreeSurfer-compatible label IDs**
Brain-Stem	16
Thalamus	10, 49
Cerebellum-Cortex	8, 47
Cerebral-White-Matter	2, 41
Cerebellum-White-Matter	7, 46
Putamen	12, 51
VentralDC	28, 60
Pallidum	13, 52
Caudate	11, 50
Lateral-Ventricle	4, 43
Hippocampus	17, 53
3rd-Ventricle	14
4th-Ventricle	15
Amygdala	18, 54
Cerebral-Cortex	3, 42
CSF	24
choroid-plexus	31, 63